GATE ME 90 Days Challenge

Day 80/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Applied Mechanics & Design	**Formula Sprint: Mechanics & Design**	**18 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a rapid formula revision of Engineering Mechanics (SHM, moments of inertia, power on a gradient), Strength of Materials (section modulus, Mohr's circle, shear-stress distribution), Theory of Machines (gear trains with idlers, Coriolis acceleration, secondary unbalanced force), Vibrations (damped natural frequency, torsional vibration, springs in series and parallel) and Machine Design (bolt tightening torque, V-belt tensions, Wahl factor) — every solution states the key formula, so the set doubles as a revision sheet. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A particle moves in simple harmonic motion with an amplitude of 50 mm. Its maximum speed is 0.5 m/s. Its maximum acceleration is

(A) 2.5 m/s² **(B)** 10 m/s²

(C) 5 m/s² **(D)** 0.5 m/s²

Q2 MCQ

A thin uniform circular disc has a mass of 4 kg and a radius of 0.2 m. Its mass moment of inertia about a tangent lying in the plane of the disc is

(A) 0.24 kg·m² **(B)** 0.20 kg·m²

(C) 0.08 kg·m² **(D)** 0.04 kg·m²

Q3 NAT

A car of mass 1200 kg climbs a road of gradient 1 in 20 (measured along the slope) at a steady 54 km/h. The road and air resistance together is 300 N. Taking g = 9.81 m/s², the power developed by the engine, in kW, is _____

Q4 MCQ

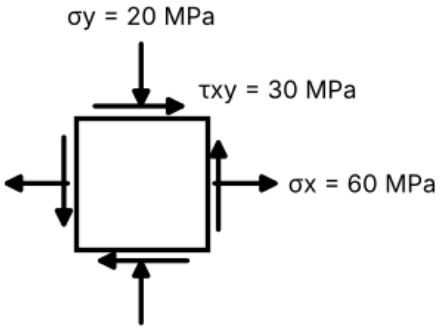
A wooden plank of cross-section 200 mm × 50 mm is used as a simply supported beam. Compared with laying it flat (50 mm vertical), placing it on edge (200 mm vertical) increases the load it can carry for the same allowable bending stress by a factor of

(A) 16 **(B)** 2

(C) 1 **(D)** 4

Q5 NAT

A plane-stress element carries σ_x = 60 MPa (tension), σ_y = 20 MPa (compression) and τ_{xy} = 30 MPa, as shown. The minimum principal stress, in MPa (with sign, tension positive), is _____



Q6 MCQ

A spring–mass–damper system has m = 5 kg, k = 20 kN/m and c = 250 N·s/m. Its damped natural frequency is

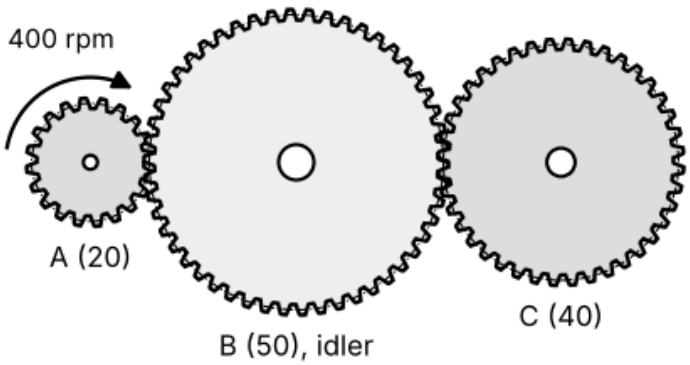
(A) 58.1 rad/s
(B) 63.2 rad/s
(C) 52.4 rad/s
(D) 9.25 rad/s

Q7 NAT

A beam of solid circular section, 100 mm in diameter, carries a shear force of 60 kN. The maximum shear stress in the section, in MPa, is _____

Q8 MCQ

In the simple gear train shown, gear A (20 teeth) drives gear C (40 teeth) through an idler B (50 teeth). If A turns at 400 rpm clockwise, gear C turns at



(A) 200 rpm counter-clockwise
(B) 200 rpm clockwise
(C) 160 rpm clockwise
(D) 500 rpm counter-clockwise

Q9 NAT

A slider moves outward along a straight rotating link with a constant velocity of 1.5 m/s relative to the link, while the link turns at a uniform 20 rad/s. The magnitude of the Coriolis component of the slider's acceleration, in m/s², is _____

Q10 MCQ

In a reciprocating engine the crank is 60 mm, the connecting rod is 240 mm and the reciprocating mass is 2 kg. The engine runs at 1500 rpm. The maximum secondary unbalanced force and its frequency are

(A) 2961 N, varying at the crank frequency
(B) 740 N, varying at the crank frequency
(C) 740 N, varying at twice the crank frequency
(D) 185 N, varying at twice the crank frequency

Q11 NAT

A rotor with a mass moment of inertia of 10 kg·m² is fixed to the free end of a vertical steel shaft 1 m long and 50 mm in diameter, whose other end is rigidly clamped. Neglecting the inertia of the shaft and taking G = 80 GPa, the natural frequency of torsional vibration, in Hz, is _____

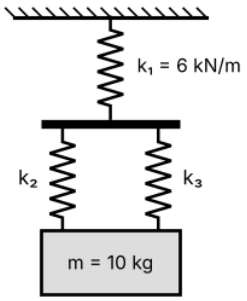
Q12 MCQ

An M16 bolt is to be tightened to an initial preload of 20 kN. Taking the usual torque coefficient of 0.2, the required tightening torque is

(A) 64 N·m
(B) 320 N·m
(C) 32 N·m
(D) 6.4 N·m

Q13 NAT

A 10 kg mass is supported as shown: spring k_1 = 6 kN/m hangs from the ceiling and carries a light rigid plate, from which the mass hangs on two springs $k_2 = k_3$ = 6 kN/m side by side. The natural frequency of vertical vibration, in Hz, is _____



$k_2 = k_3$ = 6 kN/m

Q14 MCQ

A V-belt runs in a pulley groove of included angle 40°. The angle of contact is 160° and μ = 0.25. The ratio of the tight-side to slack-side tension (ignoring centrifugal tension) is

(A) 2.01
(B) 2.96
(C) 1.27
(D) 7.70

Q15 NAT

A close-coiled helical spring has a mean coil diameter of 40 mm and a wire diameter of 5 mm. It carries an axial load of 300 N. Using Wahl's factor, the maximum shear stress in the wire, in MPa, is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(C) 5 m/s²
Q2	MCQ	(B) 0.20 kg·m²
Q3	NAT	13.33 (accept 13.2 to 13.5)
Q4	MCQ	(D) 4
Q5	NAT	−30
Q6	MCQ	(A) 58.1 rad/s
Q7	NAT	10.19 (accept 10.1 to 10.3)
Q8	MCQ	(B) 200 rpm clockwise
Q9	NAT	60
Q10	MCQ	(C) 740 N, varying at twice the crank frequency
Q11	NAT	11.15 (accept 11.05 to 11.25)
Q12	MCQ	(A) 64 N·m
Q13	NAT	3.18 (accept 3.15 to 3.21)
Q14	MCQ	(D) 7.70
Q15	NAT	289.4 (accept 286.5 to 292.3)

Detailed Solutions

Q1 **Answer: (C) 5 m/s²**

Concept: In SHM $x = A \sin \omega t$, so $v_{max} = A\omega$ and $a_{max} = A\omega^2 = v_{max}^2/A$; period $T = 2\pi/\omega$.

$\omega = v_{max}/A = 0.5/0.05 = 10$ rad/s.

$a_{max} = A\omega^2 = 0.05 \times 10^2 =$ **5 m/s² (C)**. (Period $= 2\pi/10 = 0.628$ s.)

Exam tip: Velocity is maximum at the mean position and acceleration at the extremes; 10 m/s² comes from forgetting the amplitude (using $\omega^2 \times 0.1$).

Q2 **Answer: (B) 0.20 kg·m²**

Concept: Disc: $I_{polar} = \frac{1}{2}mr^2$; $I_{diameter} = \frac{1}{4}mr^2$ (perpendicular-axis theorem); then the parallel-axis theorem $I = I_G + md^2$.

Tangent in the plane is parallel to a diameter at distance r: $I = \frac{1}{4}mr^2 + mr^2 = (5/4)mr^2$.

$I = 1.25 \times 4 \times 0.2^2 =$ **0.20 kg·m² (B)**.

Exam tip: Tangent perpendicular to the plane gives $(3/2)mr^2 = 0.24$ kg·m² — read which tangent is meant. Ring values: mr^2, $\frac{1}{2}mr^2$, $(3/2)mr^2$, $2mr^2$.

Q3 **Answer: 13.33 (accept 13.2 to 13.5)**

Concept: Power = tractive force × speed; at steady speed the tractive force = mg sinθ + resistance.

1 in 20 along the slope means sinθ = 1/20: mg sinθ = 1200 × 9.81/20 = 588.6 N.

Tractive force = 588.6 + 300 = 888.6 N; v = 54/3.6 = 15 m/s.

P = 888.6 × 15 = 13 329 W = **13.33 kW**.

Exam tip: Convert km/h to m/s (÷ 3.6) before multiplying. On an accelerating car add ma to the tractive force.

Q4 **Answer: (D) 4**

Concept: Bending strength M = σZ, with section modulus Z = bd²/6 for a rectangle (d = depth in the plane of bending).

On edge: Z = 50 × 200²/6 = 3.33 × 10^5 mm³. Flat: Z = 200 × 50²/6 = 8.33 × 10^4 mm³.

Ratio = 200/50 = **4 (D)**.

Exam tip: Strength ratio = d_1/d_2 ($Z \propto bd^2$), but the stiffness ratio is $(d_1/d_2)^2 = 16$ ($I \propto bd^3$). Do not mix them.

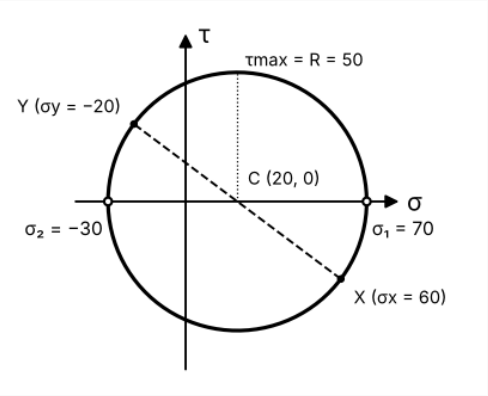
Q5 **Answer: −30**

Concept: $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2]}$; τ_{max} = radius of Mohr's circle; $\tan 2\theta_p = 2\tau_{xy}/(\sigma_x - \sigma_y)$.

Take $\sigma_y = -20$ MPa (compression is negative). Centre = (60 − 20)/2 = 20 MPa.

Radius = √(40² + 30²) = 50 MPa, so σ_1 = 70 MPa and σ_2 = 20 − 50 = **−30 MPa**; τ_{max} = 50 MPa; θ_p = ½ $\tan^{-1}$(60/80) = 18.4°.

Exam tip: The commonest error is entering the compressive stress as +20: that gives centre 40, radius 36.1 and a wrong σ_2 of 3.9 MPa.



Q6 **Answer: (A) 58.1 rad/s**

Concept: $\omega_n = \sqrt{(k/m)}$; $c_c = 2\sqrt{(km)} = 2m\omega_n$; $\zeta = c/c_c$; $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$.

$\omega_n = \sqrt{(20000/5)} = 63.25$ rad/s; $c_c = 2\sqrt{(20000 \times 5)} = 632.5$ N·s/m.

$\zeta = 250/632.5 = 0.395$; $\omega_d = 63.25 \times \sqrt{(1 - 0.156)}$ = **58.1 rad/s (A)**.

Exam tip: 52.4 rad/s is $\omega_n\sqrt{(1 - 2\zeta^2)}$, the frequency of peak amplitude in forced vibration — a different formula. 9.25 is 58.1 rad/s converted to Hz.

Q7 **Answer: 10.19 (accept 10.1 to 10.3)**

Concept: $\tau = VA\bar{y}/(Ib)$; its peak at the neutral axis is $(4/3)\tau_{avg}$ for a circle and $(3/2)\tau_{avg}$ for a rectangle.

$\tau_{avg} = V/A = 60000/(\pi/4 \times 100^2) = 7.64$ MPa.

$\tau_{max} = (4/3) \times 7.64 =$ **10.19 MPa**.

Exam tip: Thin-walled tube: $2\tau_{avg}$; I-section: the web carries nearly all the shear, $\tau_{max} \approx V/(\text{web area})$.

Q8 **Answer: (B) 200 rpm clockwise**

Concept: In a simple train, $N_{last}/N_{first} = T_{first}/T_{last}$; idlers do not change the ratio, only the direction (each external mesh reverses rotation).

$N_C = 400 \times 20/40 = 200$ rpm.

A (cw) → B (ccw) → C (cw): **200 rpm clockwise (B)**.

Exam tip: Odd number of idlers → same direction as the driver; even number → opposite. In a compound train the ratio is the product of driven/driver teeth for each pair.

Q9 **Answer: 60**

Concept: Coriolis component $a_c = 2v\omega$, where v is the sliding velocity relative to the rotating link; it acts perpendicular to the link.

$a_c = 2 \times 1.5 \times 20 =$ **60 m/s²**.

Direction: rotate the relative velocity vector by 90° in the sense of ω.

Exam tip: Do not drop the factor 2 (30 m/s² is a trap). The Coriolis term exists only when a point slides on a rotating link — never for a slider on a fixed guide.

Q10 **Answer: (C) 740 N, varying at twice the crank frequency**

Concept: Primary force = $m\omega^2 r\cos\theta$; secondary force = $m\omega^2 r\cos 2\theta/n$, where $n = l/r$; the secondary term varies at twice the crank speed.

$\omega = 2\pi \times 1500/60 = 157.1$ rad/s; $m\omega^2 r = 2 \times 157.1^2 \times 0.06 = 2961$ N; $n = 240/60 = 4$.

Secondary maximum = 2961/4 = **740 N at twice the crank frequency (C)**.

Exam tip: Dividing by n^2 (185 N) is a common slip. Think of the secondary force as from an imaginary crank of length r/(4n) rotating at 2ω.

Q11 **Answer: 11.15 (accept 11.05 to 11.25)**

Concept: Torsional stiffness $k_t = GJ/L$, $J = \pi d^4/32$; $f_n = (1/2\pi)\sqrt{(k_t/I)}$.

$J = \pi \times 0.05^4/32 = 6.136 \times 10^{-7}$ m⁴; $k_t = 80 \times 10^9 \times 6.136 \times 10^{-7}/1 = 49\,087$ N·m/rad.

$\omega_n = \sqrt{(49087/10)} = 70.06$ rad/s, $f_n = 70.06/(2\pi) =$ **11.15 Hz**.

Exam tip: The torsional formulas mirror linear ones: k → GJ/L, m → I. Including shaft inertia adds $I_{shaft}/3$ to the rotor.

Q12 **Answer: (A) 64 N·m**

Concept: Tightening torque $T = K F_i d$, with torque (nut) factor $K \approx 0.2$ for dry steel threads, d = nominal diameter.

$T = 0.2 \times 20000 \times 0.016$

$= \mathbf{64\ N{\cdot}m\ (A)}$.

Exam tip: Keep d in metres for N·m. A common design rule for preload is $F_i \approx 0.75 \times$ proof load for reusable joints (0.9 for permanent).

Q13 **Answer: 3.18 (accept 3.15 to 3.21)**

Concept: Springs in parallel: $k = k_1 + k_2$; in series: $1/k = 1/k_1 + 1/k_2$; $f_n = (1/2\pi)\sqrt{(k/m)}$.

$k_2 \parallel k_3 = 12$ kN/m; in series with k_1: $k_{eq} = (6 \times 12)/(6 + 12) = 4$ kN/m.

$\omega_n = \sqrt{(4000/10)} = 20$ rad/s; $f_n = 20/(2\pi) = \mathbf{3.18\ Hz}$.

Exam tip: Springs that carry the same force are in series; springs that share the same deflection are in parallel. Gravity does not change f_n of a linear spring–mass system.

Q14 **Answer: (D) 7.70**

Concept: Flat belt: $T_1/T_2 = e^{\mu\theta}$. V-belt: $T_1/T_2 = e^{\mu\theta\ \mathrm{cosec}\ \beta}$, where β is HALF the groove angle.

$\theta = 160° = 2.793$ rad; $\beta = 20°$: exponent $= 0.25 \times 2.793/\sin 20° = 2.041$.

$T_1/T_2 = e^{2.041} = \mathbf{7.70\ (D)}$.

Exam tip: Using the full groove angle (sin 40°) gives 2.96; forgetting the wedge effect gives 2.01. With centrifugal tension use $(T_1 - T_c)/(T_2 - T_c)$.

Q15 **Answer: 289.4 (accept 286.5 to 292.3)**

Concept: $\tau = K \times 8PD/(\pi d^3)$, with spring index $C = D/d$ and Wahl factor $K = (4C - 1)/(4C - 4) + 0.615/C$.

$C = 40/5 = 8$; $K = 31/28 + 0.615/8 = 1.107 + 0.077 = 1.184$.

$8PD/(\pi d^3) = 8 \times 300 \times 40/(\pi \times 125) = 244.5$ MPa; $\tau = 1.184 \times 244.5 = \mathbf{289.4\ MPa}$.

Exam tip: Also remember $k = Gd^4/(8D^3n)$ for stiffness. K corrects for both direct shear and curvature; the shear-stress factor alone is $K_s = 1 + 0.5/C$.

What next?

Day 80 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 81/90: Full-Length Mock 4. Comment **GATE90** on our reel to get all links.
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